Panta Rhei Research Program Executive Overview

A public research observatory for a coherent theory of reality

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DOI

EXECUTIVE SUMMARY

The program is structured enough to inspect. This is the core thesis of WP001. The Panta Rhei Research Program is not presented here as a settled theory, a substitute for review, or a finished scientific institution. It is presented as an independent open research program that has exposed its questions, construction body, result surfaces, verification routes, publications, conditional implications, and engagement interfaces in a public observatory.

The program sentence is deliberately simple: the Panta Rhei Research Program is an independent open research program dedicated to building a coherent theory of reality. WP001 explains how that program is organized, what each public lane owns, where claims should be inspected, and how serious readers can proceed without being asked for premature belief.

The live website remains the canonical, searchable superset. This white paper is the stable offline orientation route: a citable map into Program, Agenda, Corpus, Results, Verify, Publications, Impact, and Engage.

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Reader Stance and Claim Boundary

CLAIM DISCIPLINE

This Executive Overview explains how the Panta Rhei Research Program is organized as a public research observatory. It is not a primary research result, not a substitute for expert review, not a proof of T Theory, and not a claim of external acceptance for any mathematical, physical, biological, or metaphysical claim. The live website remains the canonical, searchable superset of the content. This document is an offline guide into that system.

READER ROUTE

Begin with the claim boundary, then read the overview sentence, lane map, release snapshot, and reader routes. Use the live site for current Corpus, Results, Verify, Publications, Impact, and Engage surfaces.

ROUTES AND INSPECTION

Canonical route: `https://panta-rhei.site/publications/anchor-documents/wp001-panta-rhei-research-program-executive-overview/`
Short route: `https://prrp.site/wp001`
Inspection route: `https://panta-rhei.site/verify/`
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Part I
Orientation

1. WHAT WP001 IS FOR

WP001 is the whole-program executive overview of the Panta Rhei Research Program. It answers a bounded question: what is this public research program, how is it organized, and how should a serious reader inspect it?

The answer is not a compressed theory synopsis. WP002 owns the theory synopsis. WP003 owns the TauLib technical overview. WP004 owns the reusable observatory architecture. WP005 owns the conditional public-good impact synthesis. WP001 therefore has a different task: it explains the research program as a public research observatory, with one durable orientation sentence and one practical discipline.

KEY CLAIM • Defining sentence

> The Panta Rhei Research Program is an independent open research program dedicated to building a coherent theory of reality.

The phrase "coherent theory of reality" is intentionally demanding. It does not mean an informal slogan for everything. It names a programmatic burden: a theory must explain how mathematics, physics, life, mind, and metaphysics can be read as related layers without hiding the bridges, residual assumptions, or unresolved questions that make such a project risky. The program may fail that burden. WP001 is valuable precisely because it tells the reader where to look if they want to see whether the burden is being made visible.

The website asks to be checked, not simply believed. That sentence is more than a posture. It is the design rule behind the public site. Program identity is separated from agenda obligations. Agenda obligations are separated from the Corpus build. Corpus construction is separated from Results. Results are separated from Verify. Publications are separated from claim status. Impact is separated from deployment or adoption. Engage is separated from endorsement. The separations are the inspectability of the program.

2. WHAT WP001 DOES NOT CLAIM

WP001 does not prove T Theory. It does not ask the reader to accept the program's mathematical, physical, biological, or metaphysical claims. It does not claim peer review, empirical adequacy, institutional acceptance, product availability, policy adoption, deployment readiness, or achieved public-good impact. It does not replace domain review, expert criticism, formal checking, or empirical adjudication.

It also does not turn internal program status into external scientific settlement. An internally addressed result is not external scientific acceptance. A formalized theorem is not by itself empirical truth. A citable artifact is not by itself validation. A conditional impact dossier is not by itself implementation. These distinctions are repeated throughout this document because they are not small-print disclaimers. They are structural conditions for reading the program responsibly.

The positive claim is narrower and stronger: the program is structured enough to inspect. A reader can see the lane ownership model, the release snapshot, the Corpus projections, the result catalogue, the formalization metrics, the verification surfaces, the publication shelf, the conditional impact routes, and the engagement pathways. That is not the same as correctness. It is the condition that makes serious scrutiny possible.

3. HOW TO READ THIS PAPER

Part I explains why the site is a public research observatory. Part II walks the live lane model: Discover, Program, Agenda, Corpus, Results, Verify, Publications, Impact, and Engage. Part III gives the current release snapshot and publication families. Part IV gives practical reader routes. Part V closes with the route index and citation posture.

The best way to read WP001 is to keep two windows open: this PDF and the live site. When a lane is introduced, open the corresponding site route. When a metric appears, treat it as a release snapshot, not a permanent truth. When a claim boundary appears, preserve it. The PDF is stable; the observatory is alive.

Part II

The Program as Public Research Observatory

4. WHY AN OBSERVATORY

High-scope research can fail in many ways. It can fail scientifically, by making claims that do not survive mathematical, empirical, or conceptual review. It can fail institutionally, by asking readers to trust private authority, charisma, or narrative force where inspection should be possible. It can fail editorially, by mixing orientation, construction, consequences, publication, and advocacy until no one can tell what kind of claim is being made. It can fail socially, by turning participation into endorsement.

The public observatory model is an answer to those risks. It does not solve them automatically. It makes them visible. The program places its identity, obligations, construction body, consequences, verification routes, release artifacts, conditional implications, and engagement interfaces in separate public lanes. A reader can enter through any lane, but the lanes prevent the program from pretending that one kind of evidence has already done the work of another.

In that sense the website is not merely a promotional front end. It is part of the research architecture. It says what the program is trying to do, where the burden is stated, where the construction sits, where results are classified, where verification is exposed, what has been released, what impact would mean only conditionally, and how the outside world can engage without endorsing.

5. FROM OBLIGATION TO INSPECTION

The simplest inspection chain is:

Layer	Inspection question
Program	What kind of research program is this, and what status does it claim?
Agenda	What burden has the program publicly accepted?
Corpus	What has actually been constructed, and where does it depend on other objects?
Results	What does the program currently say follows, and with what status?
Verify	What can be checked, by which method, and what remains outside that method?
Publications	What stable artifacts exist, and what kind of artifacts are they?
Impact	What would matter conditionally if upstream claims survive review?
Engage	How can readers critique, contribute, ask, or support without implying endorsement?

The chain matters because it resists a common failure mode of ambitious research communication: presenting conclusions before obligations. WP001 starts from the public program, not from the most dramatic claims. It treats the program itself as an inspectable object. A serious reader should be able to ask: what burden has been declared, what construction answers it, what results are asserted, what checks exist, and what remains open?

6. THE DOCTRINE OF SEPARATED OWNERSHIP

Separated ownership is the quiet technical doctrine of the v4 public site. Each lane can link to the others, but each lane owns a different public function. Discover may orient to Results, but it does not own result status. Program may define the research identity, but it does not discharge Agenda obligations. Corpus may ground a result, but it does not by itself validate external claims. Verify may expose formal checking, but formal checking is not empirical truth. Publications may provide a citable PDF, but artifact type is not claim status. Impact may explain conditional relevance, but impact is not deployment or adoption. Engage may invite participation, but participation does not imply endorsement.

That ownership model is why WP001 can be an executive overview without becoming a sales document. It does not ask the reader to collapse the system. It asks the reader to preserve the separations that make the system readable.

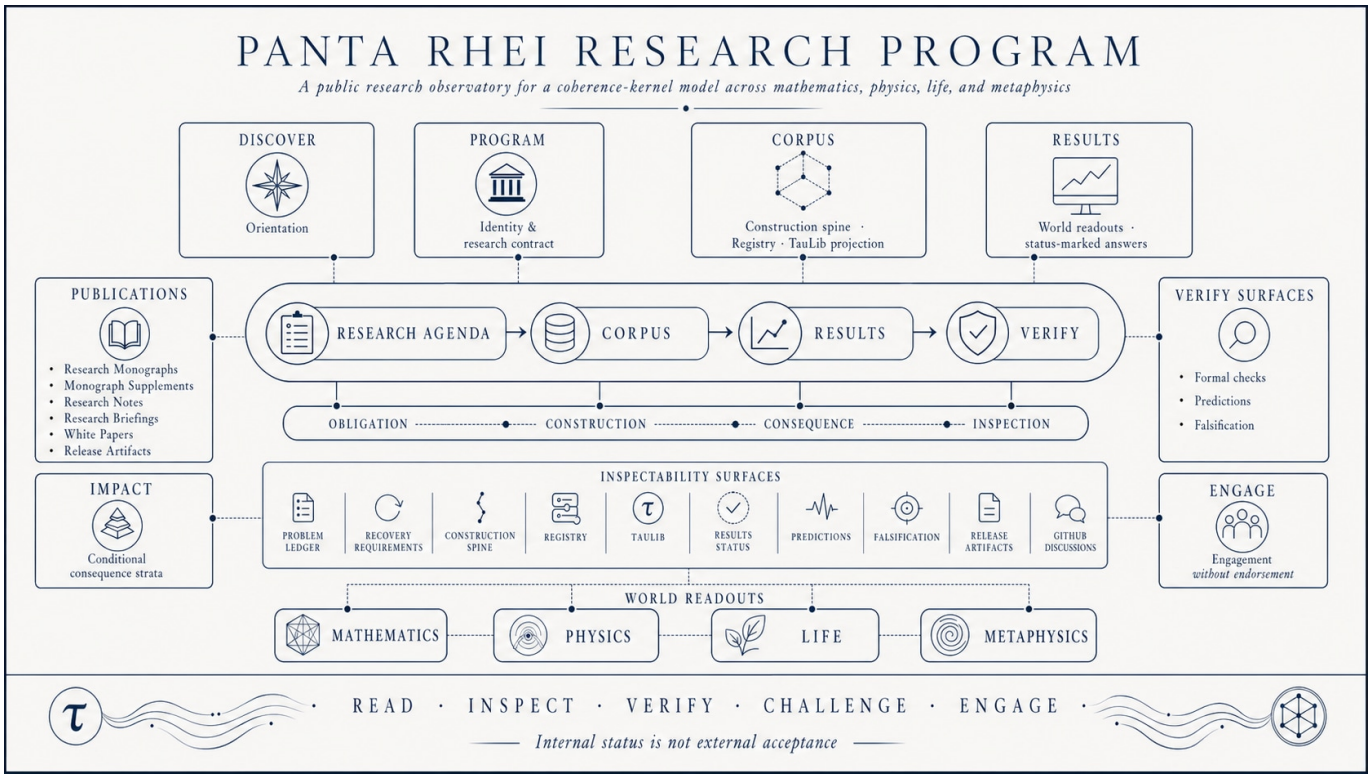

Figure 1. The public research observatory is the organizing surface of WP001: a stable orientation into live, inspectable lanes.

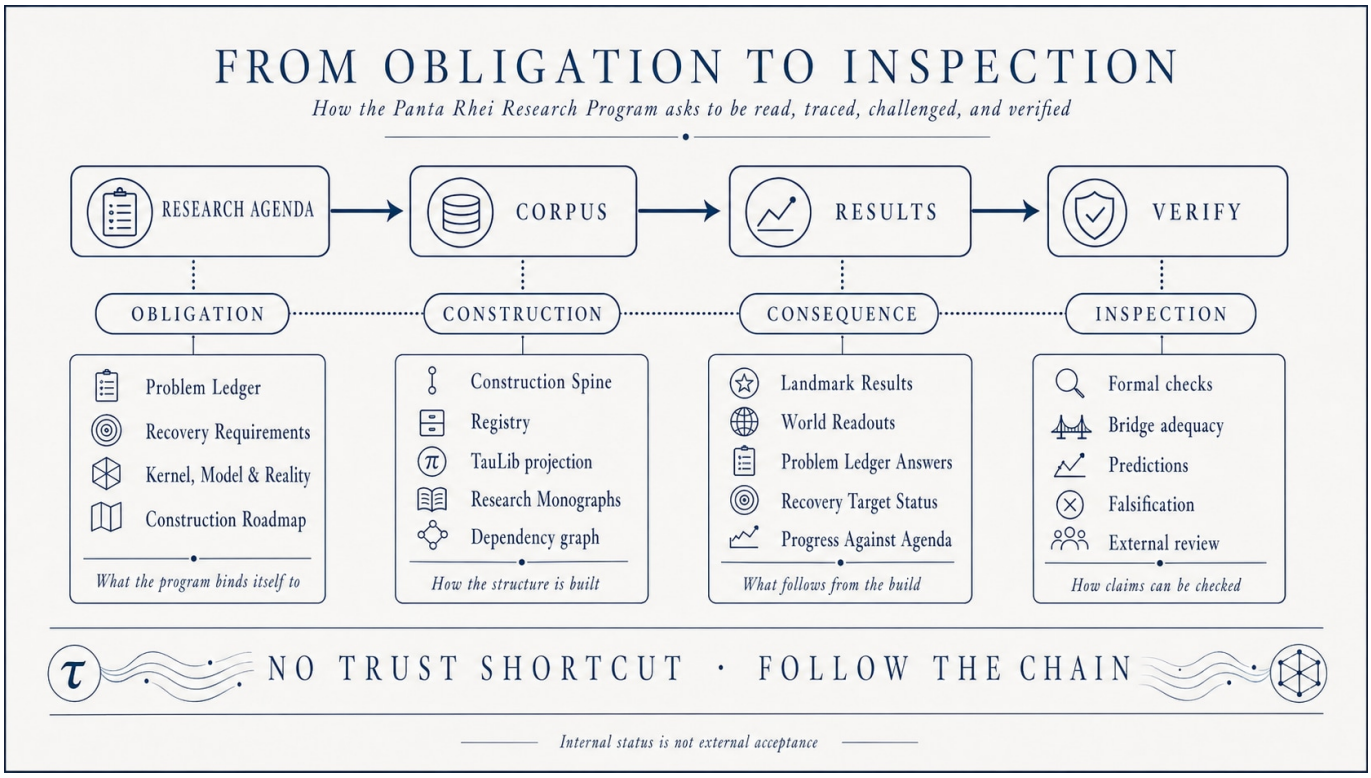

Figure 2. The observatory is designed as a route from public obligation to inspection, not as a shortcut from ambition to belief.

7. A PUBLIC SPINE, NOT A PRIVATE MAZE

The public site is also a navigation claim. A first-contact reader should be able to understand the public research program from public surfaces alone. WP001 is intentionally public in its references. It routes to website lanes, publication manifests, Corpus pages, TauLib surfaces, Results pages, Verify pages, public-good briefings, and engagement routes.

This matters for transparency. Public accountability means that public claims route to public surfaces where their status, source, and limits can be inspected. WP001 follows that rule and treats the live site as the public source of orientation.

Part III
The Lane Model I: Orientation, Identity, Obligations

8. DISCOVER: ORIENTATION BEFORE DEPTH

Discover is the first-contact layer. It does not compress the whole program into a slogan; it helps a serious reader choose the right door. The site is large because the research object is large. Discover gives orientation before depth and then routes the reader toward inspectable surfaces.

8.1 What Discover Owns

Discover owns orientation. It owns Start Here, Entry Routes, Guided Tours, Big Questions, Key Results as a first-click surface, How the System Works, publication notification guidance, and AI-assisted first-pass prompts. It helps readers decide whether they are looking for identity, obligations, construction, consequences, verification, artifacts, conditional impact, or engagement.

Discover does not own the deep content. It does not own Program doctrine, Agenda obligations, Corpus construction, Results status, Verify audit routes, Publications artifacts, Impact translation, or Engage participation. It may link to everything, but it does not own the claim status of everything it links to.

8.2 How Discover Should Be Used

A first-contact reader can begin at `/discover/`, then use `/discover/start-here/` and `/discover/how-the-system-works/`. A reader with a specific intent can use `/discover/entry-routes/`. A reader looking for high-signal examples can open `/discover/key-results/` but should continue into Results and Verify before treating those examples as settled claims.

The inspection posture is simple: Discover should teach the reader how to inspect before it tempts the reader to believe. The right output of the lane is not agreement. It is sharper questions: what obligation is being answered, where is the construction, what is the result status, what verification route exists, and what remains open?

9. PROGRAM: IDENTITY, DOCTRINE, SCOPE, SCRUTINY

Program answers what kind of research object this is. It is the identity and doctrine layer that makes Agenda, Corpus, Results, and Verify legible. It defines the program sentence, the inquiry stance, the current public status, the scope discipline, and the scrutiny posture.

9.1 The Program Sentence

KEY CLAIM • Program sentence

> The Panta Rhei Research Program is an independent open research program dedicated to building a coherent theory of reality.

The sentence has three load-bearing parts. Independent means that the work does not borrow institutional authority for claims that have not earned it. Open research means that the public observatory exposes routes for inspection, critique, correction, and review. Coherent theory of reality names the programmatic burden: a connected account across mathematical, physical, biological, mental, and metaphysical layers without hiding the bridges or residual assumptions.

9.2 What Program Owns

Program owns identity, doctrine, scope, status, scrutiny, founders, related approaches, categorical ontology, the C001 charter route, and the inspection-observatory rationale. It explains why the program uses "coherent theory of reality" rather than a loose "theory of everything" headline. It defines categorical ontology as inquiry into being through relation, transformation, context, and coherence. It states that the program is public, inspectable, canonically articulated, partially formalized, open to scrutiny, not socially settled, and not a substitute for expert review or peer review.

Program does not own the Agenda's obligations, the Corpus build, Results claims, Verify checks, Publications artifacts, or Impact portfolios. WP001 must preserve that separation. Program makes later claims legible; it does not settle them.

9.3 Program Reader Route

An institutional reader or reviewer should begin at `/program/`, then read `/program/about/` and `/program/about/scope-status-and-scrutiny/` before touching major claims. The doctrine definition belongs at `/program/about/coherent-theory-of-reality/`. The public architecture route belongs at `/program/about/inspection-observatory/`. The charter route is `/program/about/standing-in-the-inquiry-of-being/`, with C001 as the citable anchor document.

10. AGENDA: OBLIGATIONS BEFORE RESULTS

Agenda is the public obligation layer. It states what must be asked, carried, built, refused, answered, and left open before Results are allowed to sound like consequences. In short: Agenda states the burden.

That burden is necessary because a high-scope theory cannot ask for serious engagement unless it says what it owes. A coherent theory of reality must earn its language, earn its questions, build its answers, disclose its limits, and state conditional relevance before it asks for belief.

10.1 What Agenda Owns

Agenda owns Core Semantics, the Structural Challenge Ledger, Kernel, Model and Reality, Construction Roadmap, result criteria, source policy, foundational discipline, core design principles, and refusals. Core Semantics names the language the theory must earn before it can answer. The Structural Challenge Ledger records canonical stress tests the program agrees to keep visible, classify, address, reframe, or refuse with reason. Kernel, Model and Reality states the ontic-status and no-externalities burden: no hidden runtime, substrate, bridge, semantic load, or residual boundary may be treated as solved by silence. Construction Roadmap states logical build order, not a sprint plan.

Agenda does not report current results. It does not claim that named challenges have been solved. It does not own the Corpus-side construction, Results-side answer status, or Verify-side audit route. Its job is to make obligations inspectable.

10.2 Agenda Reader Route

The lane root is `/agenda/`. A reviewer should read `/agenda/core-semantics/`, then the Structural Challenge Ledger and its source policy, then `/agenda/kernel-model-reality/`, then `/agenda/construction-roadmap/`. Only after that should the reader move to Results and Verify. The route keeps obligations visible before claims are evaluated.

Part IV
The Lane Model II: Construction, Consequences, Inspection

11. CORPUS: THE CONSTRUCTION BODY

Corpus is where the theory is built. It is the construction body of the program: definitions, lemmas, theorems, structures, monograph projections, registry objects, dependency relations, and formalization pointers. Its first public question is not "what result should I believe?" but "what has been constructed, where does it sit, and what does it depend on?"

11.1 What Corpus Owns

Corpus owns the Construction Spine, the Monograph Corpus, the Registry, the Corpus Graph, foundational-hinge review routes, object types, dependency structure, and versioning posture. It owns the stable research body from which downstream claims are read.

Corpus does not own external validation, empirical acceptance, public-good impact, or final verification. It is also not the proof assistant itself. TauLib is routed through Corpus and Verify, but the proof-checking posture belongs to Verify. Publications owns citable artifacts and release shelves; Results owns answer surfaces; Verify owns inspection and challenge.

11.2 How to Inspect Corpus

The primary route is `/corpus/`. A reader should choose one of four entry points: the ten-step Construction Spine at `/corpus/construction-spine/`; the seven-book Monograph Corpus at `/corpus/monograph-corpus/`; the Registry at `/corpus/registry/`; or the guide at `/corpus/how-to-read/`. The construction-review packet at `/corpus/foundational-hinges/` is useful for the first mathematical stress test, while `/corpus/graph/` exposes dependency relations.

Corpus should be inspected as a graph, not consumed as a slogan. Start from an object, result, construction step, or monograph chapter. Follow dependencies backward, uses forward, and formalization links where available. If a result cites Corpus support, ask whether the cited object actually bears the claimed load.

12. RESULTS: CURRENT CONSEQUENCES

Results is where the built Corpus becomes a set of current consequences. It gathers landmark results, world readouts, challenge responses, core-semantics status, additional derived results, prediction catalogues, falsification surfaces, and progress-against-agenda views. Its job is to say what the program currently claims follows from the Corpus, with status markers visible.

KEY CLAIM • Result discipline

An internally addressed result is not external scientific acceptance.

"Internally addressed" means the program has an internal answer, derivation, or structural stance. It does not mean peer review, scientific consensus, empirical closure, or independent validation. That distinction is one of the most important parts of the public site.

12.1 What Results Owns

Results owns result-facing interpretation: what problem is being answered, what domain it belongs to, what kind of result it is, what status it carries, and where it routes next. It owns the public grammar of answer surfaces: internally addressed, partial, qualitative, contradicted, and not addressed.

Results does not own the underlying construction; that remains Corpus. It does not own formal proof checking, bridge adequacy, or empirical adjudication; those route to Verify. It does not own impact claims or policy adoption. Where numerical predictions or falsification packs appear in Results, they are result-facing catalogues; the accountability grammar belongs to Verify.

12.2 How to Inspect Results

The root is `/results/`. Readers can start with `/results/landmark-results/` for high-interpretive-force results, `/results/browse/` for the full catalogue, or `/results/how-to-read-a-result-page/` for result-page anatomy. Prediction and failure surfaces appear at `/results/predictions/` and `/results/falsifications/`. Progress and obligation mapping route through `/results/progress-against-agenda/` and `/results/core-semantics-status/`.

Every result should be read through four separations: result status, formal verification state, bridge adequacy, and external acceptance. A precision tier is not a status label. A status label is not peer review. A numerical agreement is not by itself a verified bridge. A contradiction label is not a mistake to hide; it is part of the accountability grammar.

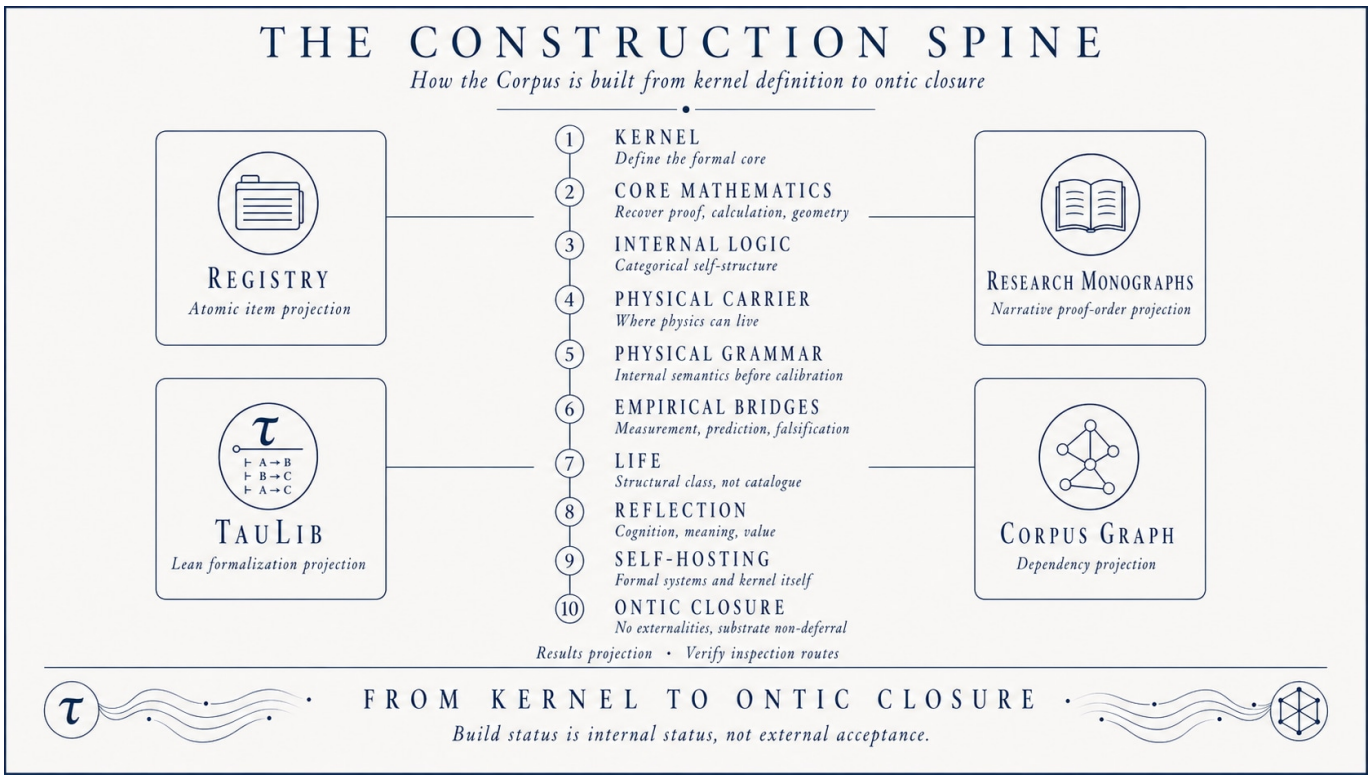

Figure 3. The Construction Spine is the human-readable route through the build order, not a replacement for the Registry or formalization surfaces.

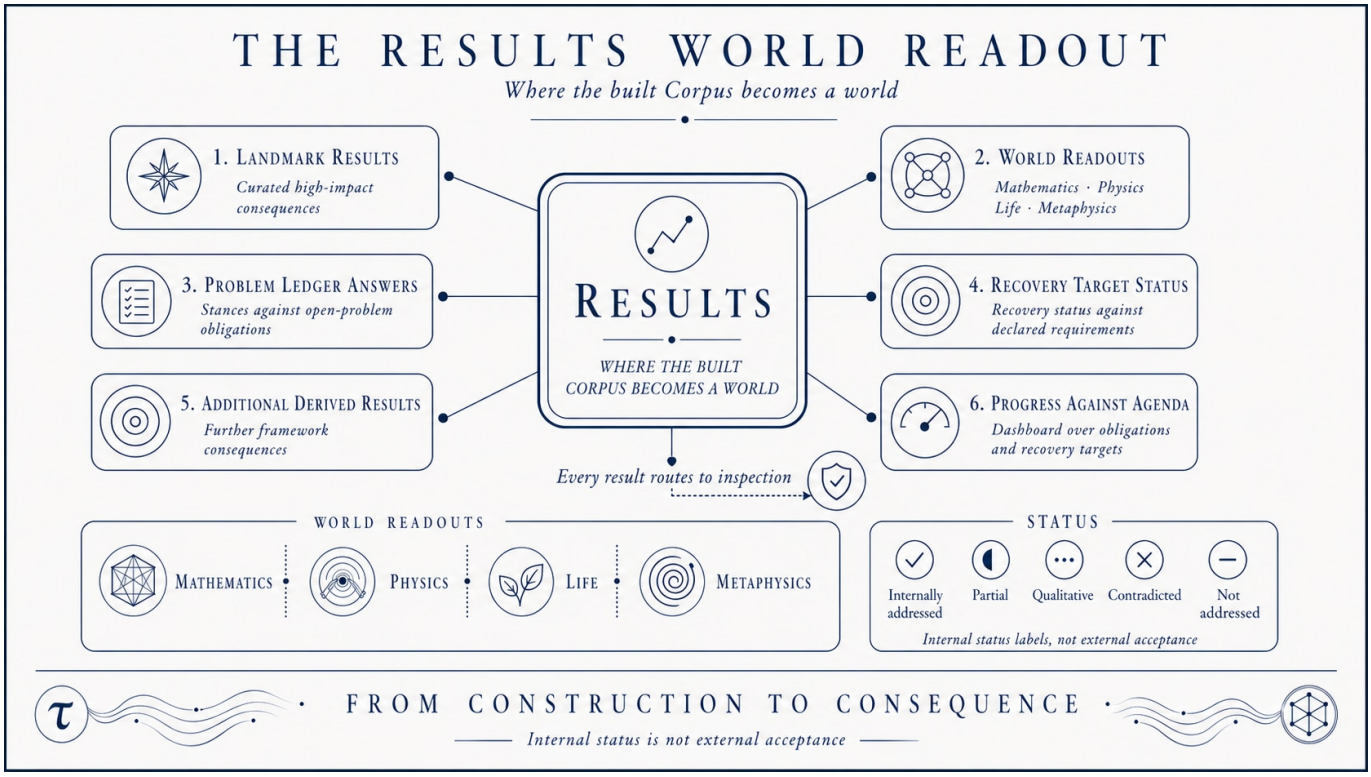

Figure 4. Results are current program consequences with visible status and routes, not claims of external settlement.

13. VERIFY: INSPECTION AND CHALLENGE

Verify is the inspection and challenge lane. It asks how obligations, construction steps, results, formal artifacts, bridges, predictions, falsification paths, and assessment protocols can be checked. Verification is not a single operation. It includes formal proof checking, bridge verification, empirical accountability, domain-specific review, trust-budget disclosure, and external assessment.

KEY CLAIM • Verification discipline

Formal checking is essential, but it is not empirical truth.

Lean compilation checks formalized internal claims relative to a trusted base and stated scope. It does not by itself establish physical truth, bridge adequacy, semantic interpretation, public-good relevance, or external scientific acceptance.

13.1 What Verify Owns

Verify owns the verification framework, scientific-rigor posture, construction-spine verification, formal verification stack, TauLib inspection bridge, release manifest, filter rules, custom axiom inventory, trusted computing base disclosure, domain-verification routes, predictions and falsification accountability, and assessment protocols.

Verify does not own the Corpus build, Results interpretation, Publications artifacts, or final public belief. Its function is to expose burdens and failure modes: what compiles, what is bridged, what remains empirical, what depends on custom axioms, what extends the trusted computing base, and what still requires external review.

13.2 Verify Reader Route

Practical entry begins at `/verify/` and `/verify/how-to-verify/`. Formal checking routes through `/verify/formal-verification-stack/`, `/verify/taulib/`, and `/verify/taulib/status/`. Bridge claims route through `/verify/bridge-verification/`. Release counts and count reconciliation route through `/verify/release-manifest/` and `/verify/filter-rules/`. Trust boundaries route through `/verify/custom-axioms/` and `/verify/tcb/`. Empirical accountability routes through `/verify/predictions-and-falsification/`.

The intended posture is neither deference nor dismissal. Start from an obligation, construction step, result, TauLib module, prediction, or assessment protocol. Name the claim type. Identify its scope. Trace it to Corpus support. Check whether a formalization exists. Ask whether the bridge from internal object to external formulation is adequate. Then ask what empirical, interpretive, or external-review burden remains.

Part V
The Lane Model III: Artifacts, Conditional Relevance, Participation

14. PUBLICATIONS: STABLE ARTIFACTS

Publications is the stable artifact layer. It gathers released objects that can be read, cited, downloaded, archived, compared across editions, and routed back into the live observatory. It is not the whole research system. Corpus carries the structured research body; Results reports current program stances; Verify exposes inspection routes; Impact maps conditional consequences. Publications provides the stable shelf of artifacts connected to those surfaces.

KEY CLAIM • Publication discipline

Artifact type is not claim status.

A monograph, briefing, white paper, note, or release manifest may be citable without thereby being externally verified, peer accepted, empirically established, or domain adopted. Publications answers a more precise question: what has been released, in what form, under what category, and through which public route can it be inspected?

14.1 Publication Families

The lane distinguishes several publication families. Anchor Documents are the citable offline canon: one charter essay and five canonical white papers that route readers back into Program, Corpus, TauLib, Publications, Impact, and Verify. Research Monographs are full book-length canonical releases of the framework, while the Monograph Corpus exposes the same deep reading order as a live Book–Part–Chapter projection. Monograph Supplements hold ledgers, appendices, derivation packs, and canonical extensions released separately from the main volumes.

Research Papers carry primary technical research contributions. Research Notes are shorter scholarly artifacts from the ongoing research stream, including comparisons, responses, pre-registration notes, falsification commitments, and current program stances. Research Briefings are translation and synthesis artifacts for particular audiences or domains. White Papers explain framework, infrastructure, method, formalization, verification architecture, or public research design. Release Artifacts document version status, provenance, corrections, manifests, changelogs, and archive state.

14.2 Anchor Documents

The Anchor Document Canon gives first-contact readers a small, citable set of offline routes into a much larger live system. C001 states the inquiry posture. WP001 orients the public research program. WP002 belongs to the theory synopsis. WP003 belongs to TauLib and the formalization surface. WP004 belongs to the public observatory architecture. WP005 belongs to conditional public-good impact.

WP001 should therefore not behave as a publication catalogue or as a compressed proof of the theory. Its role is to explain how the program is organized for inspection, and then to point readers toward the appropriate artifact family for deeper work.

15. IMPACT: CONDITIONAL RELEVANCE

Impact is the conditional consequence layer. It asks what could matter if relevant results remain supported through inspection, correction, verification, translation, domain review, and uptake. It does not turn possible implications into promises. It does not claim that the construction has already transformed science, education, institutions, public reason, or planetary systems.

KEY CLAIM • Impact discipline

Impact is conditional; it is not deployment or adoption.

The word "if" is load-bearing. No impact claim is stronger than the results, verification status, translation assumptions, domain constraints, and uptake conditions on which it depends. For that reason, Impact belongs downstream of Results and Verify. It explains why the program may matter if the construction continues to withstand scrutiny; it does not certify that such scrutiny has already been completed.

15.1 Impact Strata

The Impact lane organizes conditional relevance across several strata. Foundational Science asks what would change for the foundations of inquiry if the construction remains supported. Applied Science and Research asks how active fields could change if bridges to observation, measurement, and computation survive review and domain testing. Global Education asks how difficult domains might become teachable as related layers of one coherent construction. Existential Orientation asks what becomes thinkable if the world-picture holds. Societal Coherence asks how public reason might change if a coherence-based scientific image becomes available. Global Public Good gathers conditional portfolios where verified results could become relevant to planetary and public-good domains.

15.2 Global Public Good

Global Public Good organizes conditional portfolios for domains such as agriculture, water, climate, health, energy, biodiversity, disaster resilience, pollution, solar systems, ocean systems, and weather-facing operations. The associated Public-Good Briefings

are publication artifacts. They are Research Briefings, not deployment claims. They do not assert policy implementation, operational readiness, field validation, public-good delivery, or institutional adoption.

16. ENGAGE: SCRUTINY WITHOUT ENDORSEMENT

Engage is the participation interface of the public research observatory. It is where openness becomes operational. The program does not ask first for agreement. It asks for structured attention: careful reading, public questions, critique, reproducibility checks, domain review, correction, infrastructure contribution, responsible communication, and institutional dialogue.

KEY CLAIM • Engagement discipline

Participation does not imply endorsement.

Support does not imply validation. Review does not imply acceptance. The useful stance is to follow the claim, identify the support, name the weak point, and help decide what would count as correction, verification, or failure.

16.1 What the Program Asks For

Readers can begin by reading carefully across the lanes before judging isolated claims. Reviewers can inspect claims by tracing them into Corpus, Results, Verify, TauLib, release manifests, and assessment protocols. Critics can challenge weak links through source checks, falsification paths, result-status challenges, and correction routes.

Domain experts can review bounded areas: one structural challenge, one challenge response, one construction step, one result, one research note, one TauLib module, one briefing, or one site surface. Engineering contributors can improve documentation, metadata, search, templates, registries, TauLib documentation, import reports, and verification infrastructure. Communicators can help explain the work responsibly, without sensationalism and without treating internal results as external acceptance. Institutions can open dialogue around seminars, review settings, archive hosting, collaboration, structured evaluation, or support.

Part VI

Current Release Snapshot

17. WHY SNAPSHOT METRICS MATTER

Metrics in WP001 are release-snapshot metrics. They describe what the public observatory reports for this May 8, 2026 canonical v1.0 release. They are not permanent claims about the research program, and they are not substitutes for reading the surfaces that define the count model. The release manifest, filter rules, Registry count model, TauLib summaries, and Publications catalog are the relevant public routes.

The role of the numbers is not to impress by size. Their role is to show that the program has enough structure to inspect: a formalization surface, a Registry, a result catalogue, prediction and falsification registers, publication families, and impact dossiers.

18. SNAPSHOT TABLE

Surface	Snapshot	Reading note
TauLib modules	512	Lean module count in the public TauLib summary.
TauLib theorem and lemma records	4,863	Formalized theorem plus lemma records; formalized count, not truth count.
TauLib `sorry` count	0	No `sorry` assignments in the release summary.
Custom axioms	3	Disclosed custom axiom declarations; see custom-axiom inventory and TCB disclosure.
Registry objects	4,547	Manifest-pinned/root registry total; distinct from the typed public spine count.
Results records	255	Current Results catalogue records across Mathematics, Physics, Life, and Metaphysics.
Predictions	67	Prediction records; accountability belongs with Verify.
Falsification tests	30	Falsification records; failure routes are part of the public grammar.
Public-good portfolios	11	Impact portfolios on the Global Public Good surface.
Public-good briefings	44	Current public-good dossier/briefing surface; legacy publication records may add archival rows elsewhere.
Research papers	9	Released research-paper artifacts in the Publications catalog.
Research notes	6	Released research-note artifacts in the Publications catalog.

19. COUNT DISCIPLINE

The Registry count is the clearest example of count discipline. WP001 uses 4,547 as the manifest-pinned/root registry total. The public typed spine may show a different count, and source inventory tooling may expose still another engineering inventory number. That is not a contradiction if the count model states what each number means. The reader should not treat a count as a magic seal. The reader should inspect the counting rule.

The same caution applies to public-good briefings. WP001 uses 44 because the Impact and Public-Good Briefings surfaces list 44 current briefings across 11 portfolios. The broader Publications catalog can contain archival or superseded records, including legacy records that should not be counted as current public-good dossier slots.

20. FORMALIZATION METRICS ARE NOT TRUTH METRICS

TauLib metrics are unusually concrete: modules, lines, declarations, theorems, lemmas, `sorry` count, and custom axioms. They are central to inspectability, but they must be interpreted correctly. A Lean build can check formal statements relative to Lean, Mathlib, the trusted computing base, and explicit assumptions. It cannot by itself establish that a formal statement is the right physical bridge, that a semantic interpretation is adequate, or that an empirical claim is true.

That is why Verify separates formal verification from bridge verification, prediction and falsification, domain review, release manifest, filter rules, custom axiom inventory, and TCB disclosure. A precise formal metric is a beginning of scrutiny, not the end of it.

21. RELEASE SOURCES

The snapshot should be checked through the live site and public catalog: `/verify/release-manifest/`, `/verify/filter-rules/`, `/verify/custom-axioms/`, `/verify/tcb/`, `/corpus/registry/`, `/results/`, `/results/predictions/`, `/results/falsifications/`, `/impact/global-public-good/`, and `/publications/`. The machine-readable public catalog in the Publications repository provides additional manifest and checksum surfaces [1].

Part VII
Artifact Families

22. THE ANCHOR DOCUMENT CANON

The Anchor Document Canon is a small set of citable offline routes into the larger public observatory:

ID	Document	Role
C001	Standing in the Inquiry of Being	Program charter: what kind of inquiry this is.
WP001	Panta Rhei Research Program Executive Overview	Whole-program observatory overview: how to inspect the public program.
WP002	T Theory Executive Synopsis	Theory construction synopsis: what the theory constructs and claims.
WP003	TauLib Technical Overview	Formalization and proof-checking route.
WP004	Public Research Observatory Blueprint	Reusable architecture for inspectable high-scope open research.
WP005	Global Public Good Impact Overview	Conditional synthesis of impact strata, portfolios, and dossiers.

This canon is deliberately layered. C001 answers what kind of inquiry stands behind the program. WP001 answers what the public program is and how to inspect it. WP002 answers what the theory constructs and claims. WP003 answers how the formalization surface works. WP004 answers how the observatory design can be understood as infrastructure. WP005 answers what conditional public relevance looks like if upstream claims survive scrutiny.

23. MONOGRAPHS AND THE LIVE CORPUS

The monographs are the long-form book artifacts of the program. The live Monograph Corpus is the public reading projection into books, parts, chapters, and construction steps. The two are related but not identical in use. A book is a stable artifact. A live Corpus surface can expose current route, dependency, registry, and verification context.

WP001 should not summarize the seven books in depth. That would push it toward WP002. Instead, it should tell the reader why the monographs matter in the observatory: they are one projection of the construction body, while the Registry, Corpus Graph, TauLib projection, and Results/Verify surfaces provide other projections.

24. RESEARCH PAPERS, NOTES, BRIEFINGS, AND WHITE PAPERS

Research Papers are technical artifacts. Research Notes are shorter scholarly interventions, comparisons, or commitments. Research Briefings translate specific domains or audience-facing questions while preserving claim boundaries. White Papers explain program architecture, method, formalization, public observatory design, or public relevance.

The artifact family should never be read as claim status. A research paper can be technical and still await external review. A briefing can be polished and still conditional. A white paper can be canonical and still not be a proof. A release manifest can be exact about bytes and still not certify truth. This is why the Publications lane and Verify lane are both needed.

25. RELEASE ARTIFACTS

Release artifacts are the provenance layer. They include manifests, catalogues, route reports, checksums, changelogs, archive notes, and integrity metadata. Their job is narrower than scientific validation and more concrete than prose. They attest to artifact identity, byte integrity, source routes, version state, and public availability. They do not certify correctness, peer review, DOI registration, legal status, or empirical adequacy.

For an anchor document like WP001, the release path is intentionally explicit: a public artifact record in the Publications repository, a public PDF copy and landing page on the site, and public catalog/manifest entries connecting the artifact back to its canonical route.

Part VIII
Reader Routes

26. THIRTY-MINUTE FIRST-CONTACT ROUTE

A first-contact reader should begin with the program sentence: Panta Rhei is an independent open research program dedicated to building a coherent theory of reality, published as a public research observatory. From there, read the homepage or Discover orientation, then Program for identity and scope, then the lane overview: Agenda states the burden; Corpus carries the construction; Results reports current consequences; Verify exposes inspection routes; Publications holds stable artifacts; Impact maps conditional implications; Engage invites scrutiny without endorsement.

The goal of this route is not mastery. It is orientation: what kind of program this is, what it claims to be doing, and where a serious reader should look next.

27. REVIEWER ROUTE

A reviewer should begin with the claim boundary, then move from a specific claim into its Corpus location, result status, verification surface, source trail, and correction route. The reviewer should distinguish internal program status from external acceptance, and formal checking from empirical truth. If the claim concerns a domain consequence, the reviewer should also inspect translation assumptions and domain uptake conditions.

28. JOURNALIST OR INSTITUTIONAL ROUTE

A journalist or institutional reader should read Program, Why This Work Matters, Publications, Impact, and Engage before quoting isolated results. The safe story is not that the theory has been validated. The safe story is that an independent open research program has built a public observatory for a high-scope theory project and has exposed routes for inspection, critique, correction, and conditional relevance.

29. TECHNICAL ROUTE

A formalization or technical reader should move from Corpus to TauLib, Verify, Results, release manifests, and relevant research papers or notes. The question is not only whether an argument is persuasive in prose, but whether the formal trust budget, proof obligations, imported assumptions, and bridge claims are inspectable.

30. PUBLIC-GOOD ROUTE

A public-good reader should begin with Impact and Global Public Good, then move to Public-Good Briefings as publication artifacts. Every briefing should be read through the impact chain: upstream Results, verification and review status, translation assumptions, domain uptake, and real-world constraints. These briefings identify possible public-good pathways; they do not claim deployment, adoption, policy implementation, or operational readiness.

31. ROUTE TABLE

Reader	Start	Inspection path
First-contact reader	`/discover/`	`/program/`, `/agenda/`, `/corpus/`, `/results/`, `/verify/`
Reviewer	Claim boundary	Specific claim, Corpus support, Results status, Verify route, correction path
Journalist	`/program/`	`/publications/`, `/impact/`, `/engage/`
Formalization reader	`/corpus/taulib/`	`/verify/taulib/`, release manifest, TCB, custom axioms
Domain expert	One bounded domain page	Result status, translation assumptions, external-domain standards, Engage route
Supporter	`/engage/`	Public routes, contact routes, support posture, no-endorsement boundary

Part IX
Conclusion

32. WHAT WP001 ESTABLISHES

WP001 establishes a public orientation, not a scientific conclusion. It says that the program has a coherent enough public architecture for serious readers to inspect it. It identifies the lane ownership model, the current release snapshot, the publication families, the anchor-document canon, and the reader routes through which scrutiny can proceed.

That may sound modest next to the program's deeper ambitions, but it is the right first claim. A high-scope theory project that cannot be inspected should not ask for trust. A high-scope theory project that can be inspected still has to survive the inspection. WP001 is the map to that work.

33. WHERE TO GO NEXT

Readers who want the inquiry posture should read C001. Readers who want the theory construction should wait for or read WP002 and then move into Corpus. Readers who want formalization should wait for or read WP003 and then move into TauLib and Verify. Readers who want the observatory architecture should wait for or read WP004. Readers who want conditional public relevance should wait for or read WP005 and then read the Global Public Good portfolios and briefings.

For now, the live site remains the canonical superset: `https://panta-rhei.site/`. This PDF is the stable route back into it.

34. FINAL CLAIM BOUNDARY

The Panta Rhei Research Program may be ambitious, but ambition is not evidence. WP001 should therefore leave the reader with a disciplined stance: do not believe because the program is large; do not dismiss because the scope is unusual; inspect the structure, follow the claims, test the routes, and keep the separations visible.

The site asks to be checked, not simply believed.

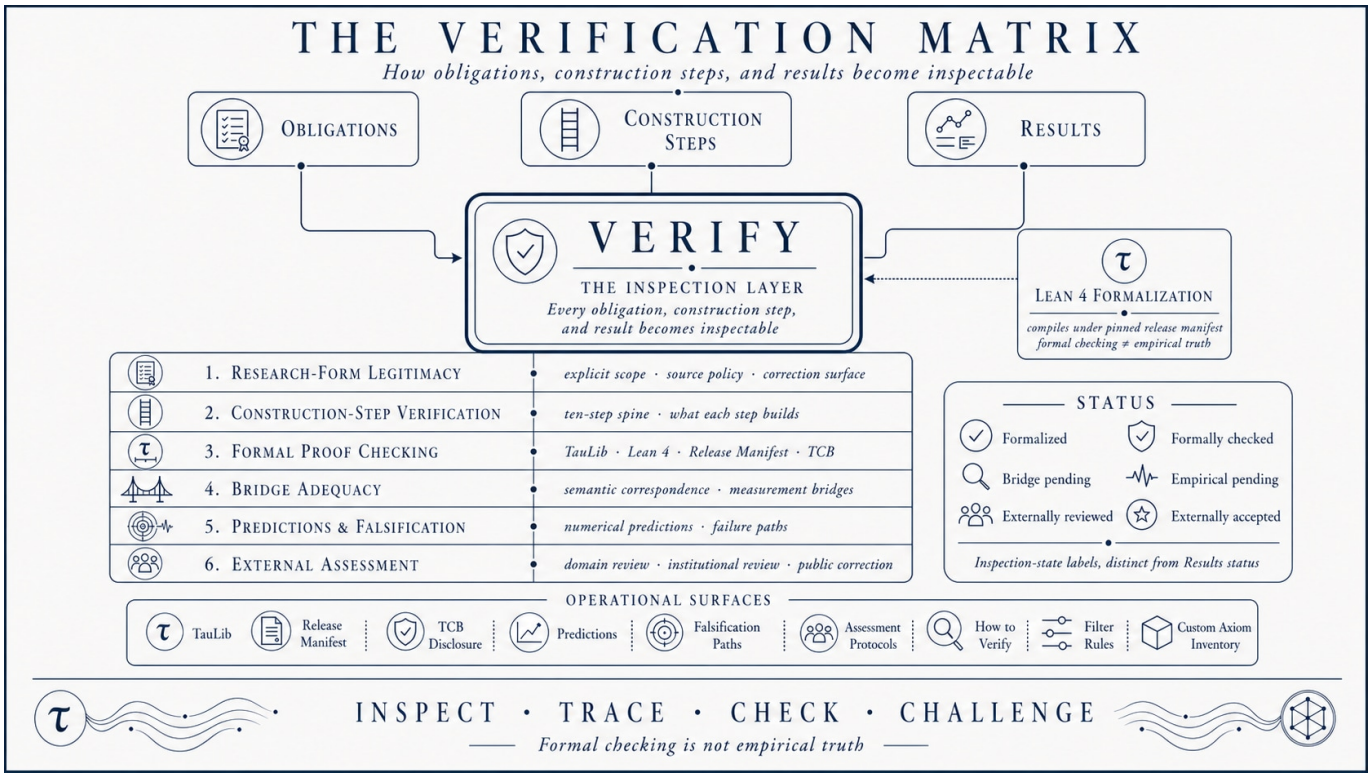

Figure 5. Verify separates formal proof checking, bridge inspection, empirical accountability, and external review rather than collapsing them into one word.

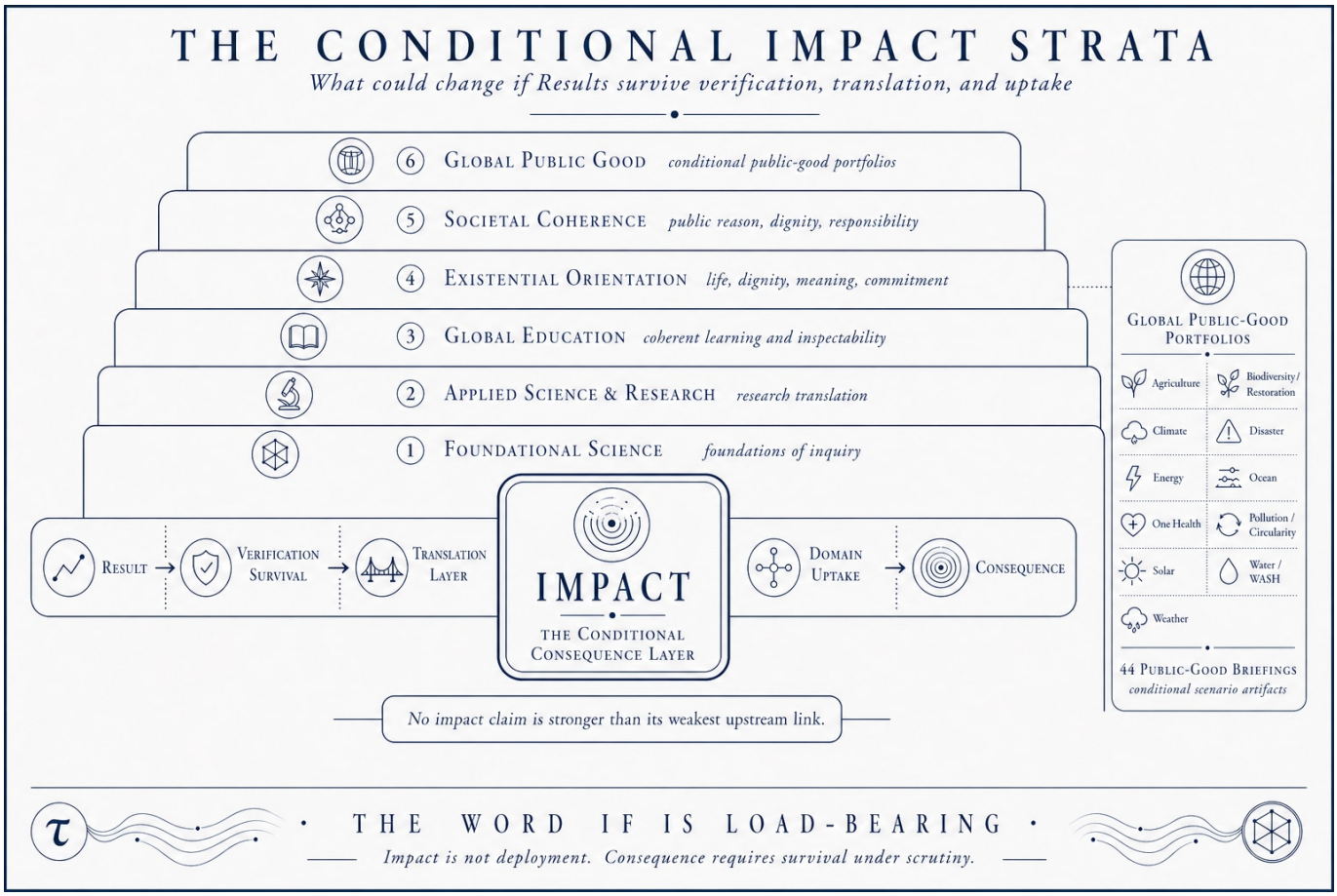

Figure 6. The Impact lane maps conditional relevance downstream of Results and Verify.

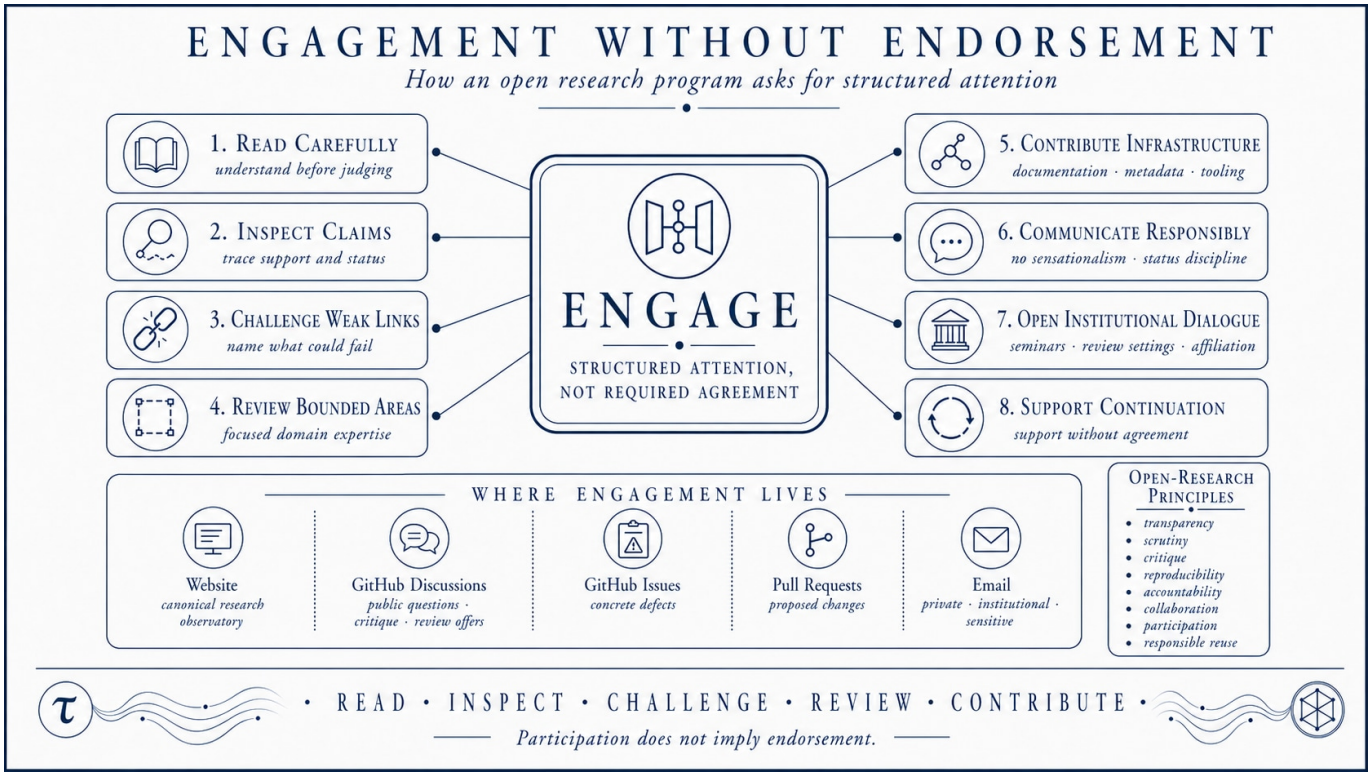

Figure 7. The Engage lane invites critique and participation without converting either into endorsement.

HOW TO CITE

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DOI:

REFERENCES

[1] Panta Rhei Research Program. Panta rhei publications catalog and manifests. https://github.com/Panta-Rhei-Research/publications, 2026. Public catalog, manifests, checksums, and route reports.

Panta Rhei Research Program
Independent open research program

WHITE PAPER

Panta Rhei Research Program Executive Overview

A public research observatory for a coherent theory of reality

WP001 • Canonical v1.0

panta-rhei.site
/publications/anchor-documents/
/engage/contact/